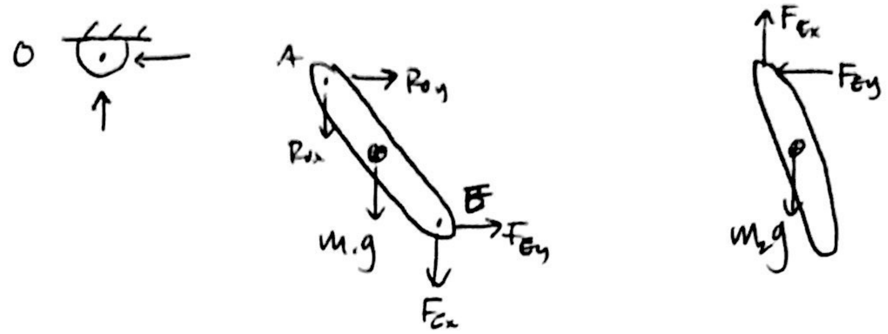
MAE 4730
10/22/2018
Lecture 24

Today:
1) Double pendulum DAEs
2) Non-holonomic constraints

DAEs:

· double pendulum FBDs



How many unknowns?
$\ddot{x}_{G1}, \ddot{y}_{G1}, \ddot{\theta}_1$
$\ddot{x}_{G2}$ $\ddot{y}_{G2}$ $\ddot{\theta}_2$
R_{ox} R_{oy} F_{Ex} F_{Ey}

How many equations?
LMB & AMB : 3
2 objects : ×2
6

constraint eqs:
2 joints × 2 DOF = 4
10

$$
\begin{array}{cccccccccc}
\ddot{x}_1 & \ddot{y}_1 & \ddot{\theta}_1 & \ddot{x}_2 & \ddot{y}_2 & \ddot{\theta}_1 & R_{ox} & R_{oy} & F_{Ex} & F_{Ey}
\end{array}
$$

$$
10\left\{
\begin{bmatrix}
m_1 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & -1 & 0 \\
0 & m_1 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & -1 \\
0 & 0 & I_1 & 0 & 0 & 0 & \multicolumn{4}{c}{\text{sines + cosines}} \\
0 & 0 & 0 & m_2 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & m_2 & 0 & 0 & 0 & 0 & 1 \\
0 & 0 & 0 & 0 & 0 & I_2 & \multicolumn{4}{c}{\text{sines + cosines}} \\
 & & & & & & 0 & 0 & 0 & 0 \\
 & & & & & & 0 & 0 & 0 & 0 \\
 & & & & & & 0 & 0 & 0 & 0 \\
 & & & & & & 0 & 0 & 0 & 0
\end{bmatrix}\right.
\begin{bmatrix}
\ddot{x}_1 \\ \ddot{y}_1 \\ \ddot{\theta}_1 \\ \ddot{x}_2 \\ \ddot{y}_2 \\ \ddot{\theta}_2 \\ R_{ox} \\ R_{oy} \\ F_{Ex} \\ F_{Ey}
\end{bmatrix}
=
\begin{bmatrix}
m_1 g \\ 0 \\ 0 \\ m_2 g \\ 0 \\ 0 \\ \hline \text{messy} \\ \text{stuff}
\end{bmatrix}
$$

Non-holonomic constraints